

0.1 Field extensions

Most fields I know have obvious prime subfields, the set of elements formed by applying all possible field operations to 1. In $\mathbb{R}$, $\mathbb{C}$, and algebraic number fields this is $\mathbb{Q}$, in finite fields this $\mathbb{F}_p$. If k is the prime field of K , there is obviously only one embeddings $k \subset K$, and for any other field $k \subset L$, all of the homomorphisms $f : K \rightarrow L$ will fix k . So it is equivalent to consider the category of fields with prime subfield k , and the category of k -extensions.

0.2 Normal extensions

Definition 0.2.1. An extension $k \subset K$ is **normal** if, for all $k \subset \Omega$, the *right* action

$$\text{Emb}_k(K, \Omega) \curvearrowright \text{Aut}_k(K)$$

by precomposition, is transitive.

This is equivalent to saying, if we have two embeddings $x_1, x_2 : L \hookrightarrow \Omega$, then there exists some automorphism $\sigma \in \text{Aut}(L/K)$ such that $x_2 = x_1\sigma$. If we just know $x_2(L) \subset x_1(L)$ as sets, this is also a field extension, and as the two fields are isomorphic they must in fact be equal in Ω . This embedding is exactly the required σ .

Lemma 0.2.2. Suppose $k \subset K$ is normal, and $k \subset F \subset K \subset \Omega$ is a tower, then the natural restriction:

$$\text{Emb}_k(K, \Omega) \hookrightarrow \text{Emb}_k(F, \Omega)$$

is surjective.

Proof. For any $x \in \text{Emb}_k(F, \Omega)$, we can form the diagram:

$$\begin{array}{ccc} K & \subset & \Omega' \\ \cup & \searrow & \cup \\ F & \xrightarrow{x} & \Omega \\ \cup & & \\ k & & \end{array}$$

where Ω' is produced by (A4), and as $k \subset K$ is normal, the inclusion $K \subset \Omega'$ factors through $K \subset \Omega$. $\square$

So when $k \subset K$ is normal, the left action $\text{Aut}(K/k) \curvearrowright \text{Emb}_k(F, K)$ by post-composition is transitive. Specifically, if $f \in k[X]$ is irreducible and splits completely in K , then $\text{Aut}(K/k)$ acts transitively on these roots, just by taking $F = k(a)$ for a some root in K .

Lemma 0.2.3. The natural composition with the given $K \subset \Omega$:

$$\text{Emb}_k(F, K) \rightarrow \text{Emb}_k(F, \Omega)$$

is also surjective.

Proof. By the previous lemma it suffices to consider $\sigma \in \text{Emb}_k(K, \Omega)$ restricted to F , but we already have a canonical $K \subset \Omega$, so σ is just represented by some element of $\text{Aut}(K/k)$ which we just restrict to F to produce our desired $\text{Emb}_k(F, K)$. $\square$

This says any k -embedding $x : F \rightarrow \Omega$ still lands F in K , this is a stronger statement of the definition of K being a normal extensions, in fact if we take $F = K$ we recover this.

Theorem 0.2.4 (Characterisation of normal extensions). For an extension $k \subset K$, the following are equivalent:

- (1) for all irreducible $f \in k[X]$, either f has no roots in K or f splits completely in K ;
- (2) there exists some $f \in k[X]$ (not necessarily irreducible) such that K is the splitting field of f over k ;
- (3) $k \subset K$ is a normal extension.

Proof. (1) $\implies$ (2) Write $K = k(a_1, \dots, a_n)$ and let f_i be the minimal polynomial of a_i over k , then K is the splitting field of $f = f_1 \cdots f_n$ over k .

(2) $\implies$ (3) For all $K \subset \Omega$ and $x \in \text{Emb}_k(K, \Omega)$, we just have to show $x(K) \subset K$. If a is a root of f in K , then so is $x(a)$ which must also be in K , as K is generated by these roots, we are done.

(3) $\implies$ (1) If f has any root $a \in K$, and we let Ω be the splitting field of f in K , we can apply the previous lemma to the tower

$$k \subset k(a) \subset K \subset \Omega$$

yielding the series of isomorphisms:

$$\{\text{roots of } f \text{ in } \Omega\} \cong \text{Emb}_k(k(a), \Omega) \cong \text{Emb}_k(k(a), K) \cong \{\text{roots of } f \text{ in } K\} \quad \square$$